AKASH KUMAR SINGH

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Professional Summary

Technical leader with extensive experience in AI/ML research and software engineering. Skilled in designing scalable systems, integrating advanced AI models, and delivering production-grade cloud solutions. Adept at guiding teams, optimizing workflows, and transforming complex challenges into reliable, high-impact products.

Technical Skill Set

- **Programming Language**: Python
- **Databases and Data Warehouse:** MySQL, MongoDB, Amazon Redshift, Redis
- **AI and Machine Learning:** Generative AI, Agentic AI, RAG, Large Language Models (LLM), Model Context Protocol (MCP), AI Model Integration, Pandas, LangChain
- **Frameworks:** FastAPI, Pydantic
- **Cloud Platform:** Amazon Web Services (AWS)
- **DevOps and Tools:** AWS Lambda, Amazon S3, Amazon SQS, Amazon CloudWatch, CodeCommit, GitLab, Argo CD
- **Project and Workflow Tools:** Jira
- **Containers:** Docker
- **IDEs:** PyCharm, Visual Studio Code
- **Operating Systems:** Linux, Windows
- **Additional Skills:** Data Engineering, ETL, RESTful APIs, Continuous Integration and Continuous Deployment (CI/CD), Analytical Skills

EDUCATIONAL QUALIFICATIONS

- **Professional Certificate in Data Science,** ICT Academy, IIT Kanpur.
- **Bachelor of Technology in Computer Science and Engineering,** Galgotias University, Delhi.
- **Senior Secondary,** B.P.S. College, Bihar.
- **Secondary,** High School Sahebganj, Bihar.

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PROFESSIONAL EXPERIENCE

Position	Team Lead (AI/ML Computational Science Specialist)
Employer	Accenture
City	Bangalore, India
Start date	Feb 2025
End date	Present
Description	• Designed and deployed AI/ML systems using Generative AI, MCP, and microservice architectures. • Implemented scalable cloud-native pipelines using Python, AWS Lambda, ECS, S3, Docker, and Kubernetes. • Integrated custom generative AI models into scientific workflows, improving automation and inference quality. • Guided engineers in Pandas, NumPy, FastAPI, MongoDB, and distributed processing practices. • Optimized ML workflows through automated preprocessing, feature engineering, and validation pipelines with CI/CD.

Position	Senior Software Engineer
Employer	Six Dee Netad Solutions Pvt Ltd (mFilterIt)
City	Bangalore, India
Start date	Aug 2019
End date	Feb 2025
Description	• Led end-to-end product development from requirement analysis to architecture, coding, testing, deployment, and maintenance. • Delivered products including Valid8, ADAO, and automated reporting systems for clients such as Amazon, Airtel, and Wakefit. • Designed and optimized high-performance RESTful APIs supporting data-driven applications. • Mentored junior developers through technical guidance, code reviews, and process improvement. • Utilized AWS services, MySQL, MongoDB, Redshift, Docker, and FastAPI to build scalable, production-ready systems.

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WORKING PROJECT

Title	CTEP (Cyber Threat Evaluation Platform)
Summary	Led the development of an enterprise-grade, AI-powered security analysis backend using Azure OpenAI, enabling advanced cyber-threat evaluation through a modular and extensible provider architecture.
Description	• Led development of an AI-powered security analysis backend using GPT-4o and Azure OpenAI. • Designed a modular provider architecture supporting multiple AI engines and rapid extensibility. • Built secure data analysis workflows for JSON, Excel, and Word files, mapping outputs to security controls. • Implemented RBAC, logging, and audit trails to meet compliance and accountability requirements.

Title	SecAI Flow
Summary	Led the architecture and development of a production-grade, multi-LLM security framework built with layered services, pluggable storage, and an agent-based orchestration engine for scalable AI-powered cybersecurity applications.
Description	• Architected and developed a scalable, multi-layer security framework powered by multiple LLMs. • Created a pluggable storage system enabling seamless switching between in-memory and PostgreSQL backends. • Built a thread-safe Unified Storage Service to support reliable data operations. • Developed an enterprise-grade FastAPI gateway with authentication, rate limiting, streaming responses, and audit logging. • Added full test coverage and documentation for production readiness.

Title	Valid8
Summary	Led a team of 6 members in developing scalable applications using Python and AWS services.
Description	• Led a team of six engineers to develop scalable applications using Python and AWS services. • Migrated legacy systems to AWS, reducing operational costs. • Built data solutions using MySQL and MongoDB for analytical workloads. • Conducted code reviews and performance evaluations to improve team output. • Developed RESTful APIs using FastAPI for a high-performance backend service. • Used Pandas for data transformation, analysis, and pipeline development in data engineering workflows.